ON-BOARD DIAGNOSTIC SYSTEM [BODY CONTROL MODULE (BCM)]

id094000050100

Outline

- The on-board diagnostic function consists of the following functions:
  - — A malfunction detection function, which detects overall malfunctions in the BCM-related parts.
  - — A memory function, which stores detected DTCs.
  - — A display function, which indicates malfunction locations and status via DTC output.
  - — A PID/data monitoring function, which reads out specific input/output signals and verifies the input/output condition.
- Using the Mazda Modular Diagnostic System (M-MDS), DTCs can be read out and deleted, and the PID/data monitoring function can be activated.

Malfunction detection function

- Detects malfunctions in input/output signals.
- If a malfunction occurs, the BCM records the malfunction as a DTC. A recorded DTC can be read by the Mazda Modular Diagnostic System (M-MDS).

DTC table

×: Applicable
—: Not applicable

DTC No.	Warning/ indicator light	Description	Fail-safe	Drive cycle	Self test type*1	Memory function
B1048:7B	—	Brake fluid level sensor circuit malfunction	—	—	D	×
B1079:13*2	—	Bonnet latch switch circuit malfunction	—	—	D	×
B108F:16*3	—	Door lock switch circuit malfunction	—	—	C, D	×
B109E:87	—	Communication error with start stop unit	—	—	C, D	×
B109F:49*4	—	Intruder sensor internal malfunction	—	—	C, D	×
B109F:87*4	—	Communication error with intruder sensor	—	—	C, D	×
B10A0:64	—	Wiper and washer switch error signal received	×	—	C, D	×
B10A5:49*4	—	Theft-deterrent siren internal malfunction	—	—	C, D	×
B10A5:87*4	—	Communication error with theft-deterrent siren	—	—	C, D	×
B10A6:62	—	Light switch position signal error	×	—	C, D	×
B10A6:64	—	Light switch error signal received	×	—	C, D	×
B10AD:49*5	—	Auto light sensor/rain sensor internal malfunction	×	—	C, D	×
B10AD:56*5	—	Auto light sensitivity mismatch	×	—	C, D	×
B10AD:86*5	—	Auto wiper control request signal error	×	—	C, D	×
B113E:11	—	**4SD** • Trunk lid opener switch (exterior) circuit malfunction **5HB** • Liftgate opener switch circuit malfunction	—	—	D	×
B1172:11	—	Front door lock-link switch (driver's side) unlock circuit malfunction	—	—	D	×
B1172:13	—	Front door lock-link switch (driver's side) unlock circuit malfunction	—	—	D	×
B1174:11*6	—	Door lock-link switch unlock circuit malfunction	—	—	C, D	×

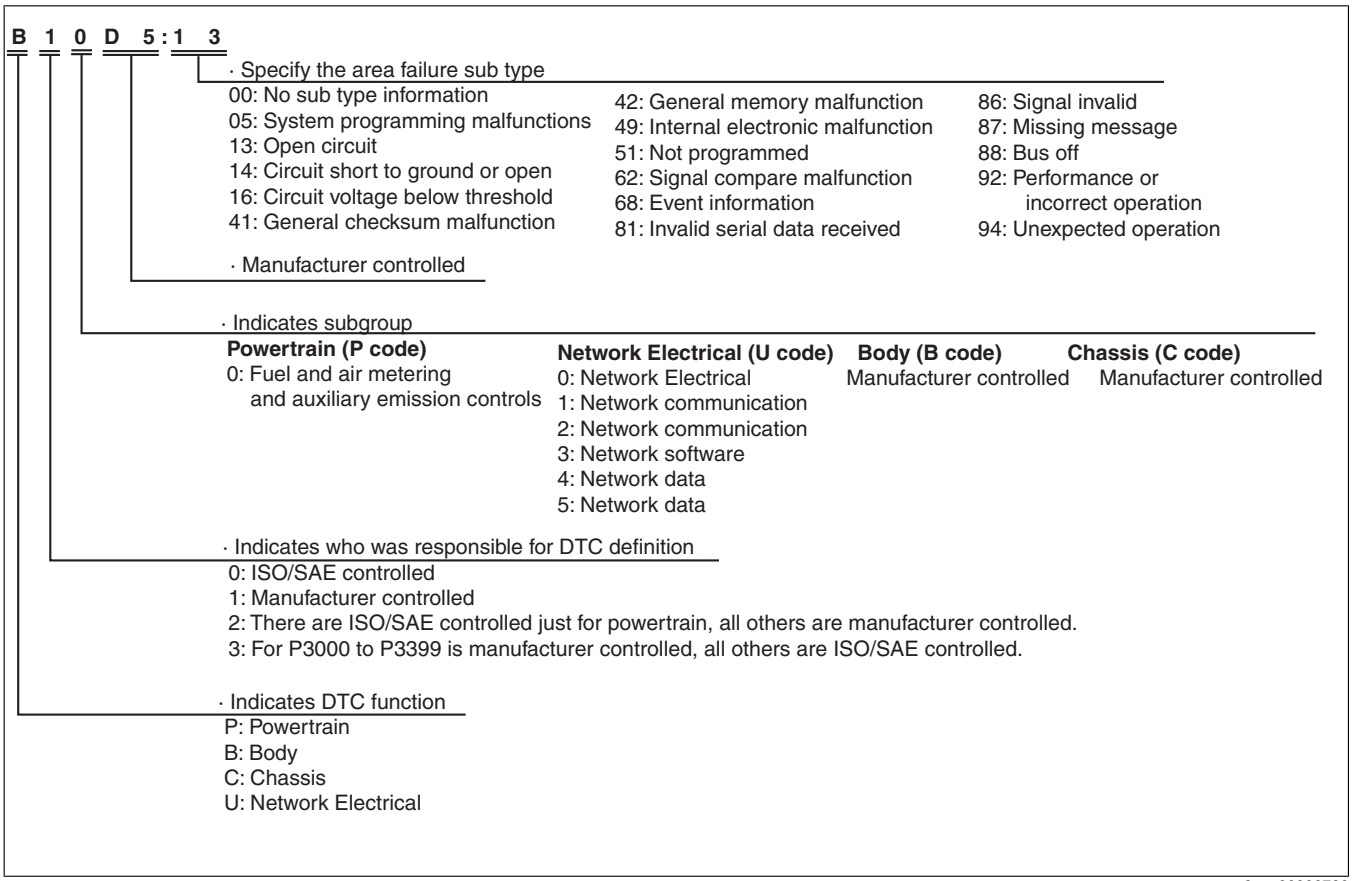
DTC No.	Warning/ indicator light	Description	Fail-safe	Drive cycle	Self test type*1	Memory function
B1174:13*6	—	Door lock-link switch unlock circuit malfunction	—	—	C, D	×
B1175:11	—	Front door latch switch (driver's side) circuit malfunction	—	—	D	×
B1176:11	—	Front door latch switch (passenger's side) circuit malfunction	—	—	D	×
B1178:11	—	**4SD** • Trunk lid latch switch circuit malfunction **5HB** • Liftgate latch switch circuit malfunction	—	—	D	×
B11DA:16	—	Front door key cylinder switch (driver's side) circuit malfunction	—	—	D	×
B11DB:86	—	Error signal received from current sensor	—	—	C, D	×
B11E9:11	—	Rear door latch switch (LH) circuit malfunction	—	—	D	×
B11EA:11	—	Rear door latch switch (RH) circuit malfunction	—	—	D	×
B1219:11*7	—	Trunk lid opener switch (interior) circuit malfunction	—	—	D	×
B126A:11	—	Front door lock-link switch (driver's side) lock circuit malfunction	—	—	D	×
B126A:13	—	Front door lock-link switch (driver's side) lock circuit malfunction	—	—	D	×
B12D3:11	—	Front door lock-link switch (driver's side) circuit malfunction	—	—	C, D	×
B1D06:11	—	Turn light (LH) circuit malfunction	—	—	C, D	×
B1D06:13	—	Turn light (LH) circuit malfunction	—	—	C, D	×
B1D07:11	—	Turn light (RH) circuit malfunction	—	—	C, D	×
B1D07:13	—	Turn light (RH) circuit malfunction	—	—	C, D	×
B1D35:86	—	Hazard warning switch error signal received	×	—	C, D	×
B1D36:86	—	Turn switch error signal received	×	—	C, D	×
C0023:14*8	—	Brake light circuit malfunction	×	—	C, D	×
C1137:86	—	Back-up light on request error signal received	×	—	C, D	×
U0001:88	—	Module communication error (HS-CAN)	×	—	C, D	×
U0100:00	—	Communication error with PCM	×	—	C, D	×
U0101:00*9	—	Communication error with TCM	—	—	C, D	×
U0121:00	—	Communication error with ABS HU/CM or DSC HU/CM	—	—	C, D	×
U0142:68	—	Communication error with electrical supply unit (ESU)	—	—	C, D	×
U0151:00	—	Communication error with SAS control module	—	—	C, D	×
U0155:00	—	Communication error with instrument cluster	×	—	C, D	×
U0214:00	—	Communication error with start stop unit	×	—	C, D	×
U0231:68*5	—	Communication error with auto light sensor/rain sensor	×	—	C, D	×
U023A:00*10	—	Communication error with forward sensing camera (FSC)	—	—	C, D	×

DTC No.	Warning/ indicator light	Description	Fail-safe	Drive cycle	Self test type*1	Memory function
U0298:68* 11	—	Communication error with DC-DC converter	—	—	C, D	×
U0452:68	—	Fuel-cut error signal at time of collision received	—	—	C, D	×
U0515:41	—	Error signal received from start stop unit	×	—	C, D	×
U053B: 68*10	—	Error signal received from forward sensing camera (FSC)	×	—	C, D	×
U0599:86* 11	—	Error signal received from DC-DC converter	—	—	C, D	×
U1007:68	—	Communication error with current sensor	—	—	C, D	×
U2300:54	—	Configuration data not received from instrument cluster	—	—	C, D	×
U2300:55	—	Instrument cluster configuration not implemented	—	—	C, D	×
U2300:56	—	Configuration data unmatched with instrument cluster	—	—	C, D	×
U3000:44	—	BCM internal malfunction	×	—	C, D	×
U3000:49	—	BCM internal malfunction	×	—	C, D	×
U3003:16	—	BCM power supply voltage low input	—	—	C, D	×
U3003:17	—	BCM power supply voltage high input	—	—	C, D	×

*1 : C: CMDTC self test, D: ODDTC self test
*2 : With bonnet latch switch
*3 : With door lock switch
*4 : With intruder sensor
*5 : With auto light/auto wiper system
*6 : With theft-deterrent system
*7 : With trunk lid opener switch (interior)
*8 : With smart city brake support (SCBS)
*9 : ATX
*10 : With forward sensing camera (FSC)
*11 : With i-stop, without i-ELOOP

DTC 7-digit code definition

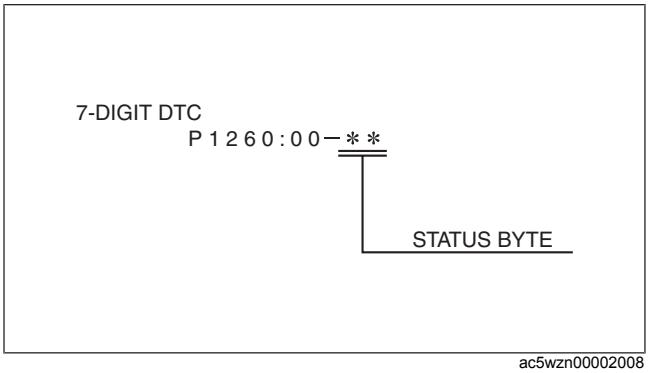
- When related systems or components have failed, the CM stores the DTC of the malfunctioning part in the CM memory, and allows for the retrieval of the store data using scanning tool when necessary. The DTCs are indicated using seven digits. Each digit indicates the following.



am6xun00000789

Status byte for DTC

- The two digits (two digits after hyphen (-)) after the 7-digit DTC.
- A code which indicates the pending code, current/ past malfunction status, or warning illumination status.
- Can be read by performing a CMDTC self-test using the Mazda Modular Diagnostic System (M-MDS).
- For details on the status byte, refer to the explanation on the Mazda Modular Diagnostic System (M-MDS) when reading the DTC.



ac5wzn00002008

Detection condition for the applicable DTC

DTC No.	Description	Detection condition
B1048:7B	Brake fluid level sensor circuit malfunction	• BCM detected that the brake fluid level is less than MIN with the ignition switched ON (engine off or on).
B1079:13	Bonnet latch switch circuit malfunction	• BCM detects an open circuit in bonnet latch switch circuit with the ignition switched ON (engine off or on).
B108F:16	Door lock switch circuit malfunction	• With the door lock switch off, the BCM detects a short to ground in the door lock switch circuit.
B109E:87	Communication error with start stop unit	• BCM detects a communication error with the start stop unit for **10 times continuously** with the ignition switched ON (engine off or on).
B109F:49	Intruder sensor internal malfunction	• BCM detects the intruder sensor internal malfunction for **1.5 s or more**.
B109F:87	Communication error with intruder sensor	• BCM cannot receive the signal from the intruder sensor for **1.5 s or more** with the ignition switched ON (engine off or on).
B10A0:64	Wiper and washer switch error signal received	• BCM receives error signals from the start stop unit for **5 s or more** with the ignition switched ON (engine off or on).

DTC No.	Description	Detection condition
B10A5:49	Theft-deterrent siren internal malfunction	• BCM detects the theft-deterrent siren internal malfunction for **1.5 s or more**.
B10A5:87	Communication error with theft-deterrent siren	• BCM cannot receive the signal from the theft-deterrent siren for **1.5 s or more** with the ignition switched ON (engine off or on).
B10A6:62	Light switch position signal error	• The BCM detected a light switch position signal error received from the start stop unit for **5 s or more** with the ignition switched ON (engine off or on).
B10A6:64	Light switch error signal received	• BCM receives error signals from the start stop unit for **5 s or more** with the ignition switched ON (engine off or on).
B10AD:49	Auto light sensor/rain sensor internal malfunction	• The BCM received auto light sensor/rain sensor error signals from the auto light sensor/rain sensor **ten times continuously** with the ignition switched ON (engine off or on).
B10AD:56	Auto light sensitivity mismatch	• The BCM received the following mismatches for **5 s or more** with the ignition switched ON (engine off or on). — Auto light sensitivity setting required by BCM to set auto light sensor/rain sensor — Current value of auto light sensitivity setting which auto light sensor/rain sensor sends to BCM
B10AD:86	Auto wiper control request signal error	• BCM receives the error signals from the auto light sensor/rain sensor for **5 s or more** with the ignition switched ON (engine off or on).
B113E:11	**4SD** • Trunk lid opener switch (exterior) circuit malfunction **5HB** • Liftgate opener switch circuit malfunction	**4SD** • BCM detects a short to ground in the trunk lid opener switch (exterior) circuit. **5HB** • BCM detects a short to ground in the liftgate opener switch circuit.
B1172:11	Front door lock-link switch (driver's side) unlock circuit malfunction	• BCM detects a short to ground in the front door lock-link switch (driver's side) unlock circuit with the front door lock-link switch (driver's side) locked.
B1172:13	Front door lock-link switch (driver's side) unlock circuit malfunction	• BCM detects an open circuit in the front door lock-link switch (driver's side) unlock circuit with the front door lock-link switch (driver's side) unlocked.
B1174:11	Door lock-link switch unlock circuit malfunction	• BCM detects a short to ground in the door lock-link switch unlock circuit with the front door lock-link switch (passenger's side)/rear door lock-link switch (LH)/(RH) not unlocked.
B1174:13	Door lock-link switch unlock circuit malfunction	• BCM detects an open circuit in the door lock-link switch unlock circuit with the front door lock-link switch (passenger's side)/rear door lock-link switch (LH)/(RH) unlocked.
B1175:11	Front door latch switch (driver's side) circuit malfunction	• BCM detects a short to ground in the front door latch switch (driver's side) circuit with the front door (driver's side) closed (front door latch switch (driver's side) off).
B1176:11	Front door latch switch (passenger's side) circuit malfunction	• BCM detects a short to ground in the front door latch switch (passenger's side) circuit with the front door (passenger's side) closed (front door latch switch (passenger's side) off).
B1178:11	**4SD** • Trunk lid latch switch circuit malfunction **5HB** • Liftgate latch switch circuit malfunction	**4SD** • BCM detects short to ground in trunk lid latch switch circuit with the trunk closed (trunk lid latch switch off). **5HB** • BCM detects short to ground in liftgate latch switch circuit with the liftgate closed (liftgate latch switch off).
B11DA:16	Front door key cylinder switch (driver's side) circuit malfunction	• BCM detects the front door key cylinder switch (driver's side) circuit voltage of **below 4 V** with the front door key cylinder switch (driver's side) off.
B11DB:86	Error signal received from current sensor	• BCM receives the error signals from the current sensor for **5 s or more** with the ignition switched ON (engine off or on).

DTC No.	Description	Detection condition
B11E9:11	Rear door latch switch (LH) circuit malfunction	• BCM detects a short to ground in the rear door latch switch (LH) circuit with the rear door (LH) closed (rear door latch switch (LH) off).
B11EA:11	Rear door latch switch (RH) circuit malfunction	• BCM detects a short to ground in the rear door latch switch (RH) circuit with the rear door (RH) closed (rear door latch switch (RH) off).
B1219:11	Trunk lid opener switch (interior) circuit malfunction	• BCM detects a short to ground in trunk lid opener switch (interior) circuit.
B126A:11	Front door lock-link switch (driver's side) lock circuit malfunction	• BCM detects a short to ground in the front door lock-link switch (driver's side) lock circuit with the front door lock-link switch (driver's side) unlocked.
B126A:13	Front door lock-link switch (driver's side) lock circuit malfunction	• BCM detects an open circuit in the front door lock-link switch (driver's side) lock circuit with the front door lock-link switch (driver's side) locked.
B12D3:11	Front door lock-link switch (driver's side) circuit malfunction	• Any of the following conditions is met with the ignition switched ON (engine off or on). — The BCM detected a short to ground in the front door lock-link switch (driver's side) lock side circuit for **5 s or more** with the front door lock-link switch (driver's side) unlocked. — The BCM detected a short to ground in the front door lock-link switch (driver's side) unlock circuit for **5 s or more** with the front door lock-link switch (driver's side) locked.
B1D06:11	Turn light (LH) circuit malfunction	• BCM detects a short to ground in the turn light (LH) circuit with the ignition switched ON (engine off or on).
B1D06:13	Turn light (LH) circuit malfunction	• BCM detects an open circuit or short to power supply in the turn light (LH) circuit with the ignition switched ON (engine off or on).
B1D07:11	Turn light (RH) circuit malfunction	• BCM detects a short to ground in the turn light (RH) circuit with the ignition switched ON (engine off or on).
B1D07:13	Turn light (RH) circuit malfunction	• BCM detects an open circuit or short to power supply in the turn light (RH) circuit with the ignition switched ON (engine off or on).
B1D35:86	Hazard warning switch error signal received	• BCM receives the error signals from the start stop unit for **2 s or more** with the ignition switched ON (engine off or on).
B1D36:86	Turn switch error signal received	• BCM receives the error signals from the start stop unit for **2 s or more** with the ignition switched ON (engine off or on).
C0023:14	Brake light circuit malfunction	• BCM detects any of the following conditions: — Brake lights/high-mount brake light illuminated even though driver or system does not illuminate brake lights/high-mount brake lights. — Brake lights/high-mount brake light not illuminated even though driver depresses brake pedal. — Brake lights/high-mount brake light not illuminated even though system illuminates brake lights/high-mount brake light.
C1137:86	Back-up light on request error signal received	• BCM receives the error signals from the instrument cluster for **2 s or more** with the ignition switched ON (engine off or on).
U0001:88	Module communication error (HS-CAN)	• BCM detects the CAN bus communication line (HS-CAN) malfunction **10 times** continuously with the ignition switched ON (engine off or on).
U0100:00	Communication error with PCM	• BCM could not receive the CAN signal from the PCM for 5 **s or more** with the ignition switched ON (engine off or on).
U0101:00	Communication error with TCM	• BCM could not receive the CAN signal from the TCM for **5 s or more** with the ignition switched ON (engine off or on).

DTC No.	Description	Detection condition
U0121:00	Communication error with ABS HU/CM or DSC HU/CM	• BCM could not receive the CAN signal from the ABS HU/CM or DSC HU/CM for **5 s or more** with the ignition switched ON (engine off or on).
U0142:68	Communication error with electrical supply unit (ESU)	• BCM could not receive the signals from the electrical supply unit (ESU) for **5 s or more** with the ignition switched ON (engine off or on).
U0151:00	Communication error with SAS control module	• BCM could not receive the CAN signal from the SAS control module for **5 s or more** with the ignition switched ON (engine off or on).
U0155:00	Communication error with instrument cluster	• BCM could not receive the CAN signal from the instrument cluster for **5 s or more** with the ignition switched ON (engine off or on).
U0214:00	Communication error with start stop unit	• BCM could not receive the CAN signal from the start stop unit for **5 s or more** with the ignition switched ON (engine off or on).
U0231:68	Communication error with auto light sensor/rain sensor	• BCM could not receive the signals from the auto light sensor/rain sensor for **5 s or more** with the ignition switched ON (engine off or on).
U023A:00	Communication error with forward sensing camera (FSC)	• BCM could not receive the CAN signal from the forward sensing camera (FSC) for **5 s or more** with the ignition switched ON (engine off or on).
U0298:68	Communication error with DC-DC converter	• BCM could not receive the signals from the DC-DC converter for **5 s or more** with the ignition switched ON (engine off or on).
U0452:68	Fuel-cut error signal at time of collision received	• The BCM received a fuel-cut error signal at the time of a collision from the SAS control module for **3 s or more** with the ignition switched ON (engine off or on).
U0515:41	Error signal received from start stop unit	• Check sum error occurred on communication with the BCM and the start stop unit **3 times continuously** with the ignition switched ON (engine off or on).
U053B:68	Error signal received from forward sensing camera (FSC)	• BCM receives the error signals from the forward sensing camera (FSC) for **5 s or more** with the ignition switched ON (engine off or on).
U0599:86	Error signal received from DC-DC converter	• BCM receives the error signals from the DC-DC converter for **5 s or more** with the ignition switched ON (engine off or on).
U1007:68	Communication error with current sensor	• BCM could not receive signals from the current sensor for **5 s or more** with the ignition switched ON (engine off or on).
U2300:54	Configuration data not received from instrument cluster	• BCM cannot receive the configuration data from the instrument cluster with the ignition switched ON (engine off or on).
U2300:55	Instrument cluster configuration not implemented	• The BCM does not perform the configuration with the ignition switched ON (engine off or on).
U2300:56	Configuration data unmatched with instrument cluster	• Configuration data of the BCM and instrument cluster are not matched with the ignition switched ON (engine off or on).
U3000:44	BCM internal malfunction	• BCM detects the internal malfunction with the ignition switched ON (engine off or on).
U3000:49	BCM internal malfunction	• BCM detects the internal malfunction with the ignition switched ON (engine off or on).
U3003:16	BCM power supply voltage low input	• The BCM detected a BCM power supply circuit (terminals 1A and 1B) voltage of **less than 10 V** for **10 s or more** with the ignition switched ON (engine off or on).
U3003:17	BCM power supply voltage high input	• The BCM detected the BCM power supply circuit (terminals 1A or 1B) voltage of **16 V or more** for **10 s or more** with the ignition switched ON (engine off or on).

Snapshot Data

Note
- The BCM stores the following two types of snapshot data (vehicle information) when a DTC is detected and displays them in the Mazda Modular Diagnostic System (M-MDS).
  — Vehicle information detected by BCM
  — Vehicle information detected by instrument cluster and received by BCM via CAN communication

- The data for all DTCs currently detected is stored.

—: Not applicable

Snapshot data item	Unit		Data contents	Data read/use method	Corresponding data monitor items
AAT	°C	°F	Ambient temperature	—	—
APP_STATUS	Accelerator Pedal Off/Under20%/ Over20%/FAIL		Accelerator pedal position status	—	—
CFG_STATUS	Config Complete/ Not Configured/ Config Error		Instrument cluster configuration status	—	—
ECT_STATUS	Under 0 degrees C/ 0 - Under 80 degrees C/Over 80 degrees C/FAIL		Engine coolant temperature status	—	—
IC_VPWR	V		Instrument cluster power supply voltage	• The BCM receives the power supply voltage value of the instrument cluster sent via CAN communication from the instrument cluster. • If a DTC is detected, the BCM records the power supply voltage of the instrument cluster when the DTC was detected, and it is displayed in the Mazda Modular Diagnostic System (M-MDS).	VPWR*1
IG-ON_TIMER	hh:mm:ss*2		Elapsed time since ignition was switched ON (engine off or on) **Note** • The instrument cluster records the elapsed time since the ignition was switched ON (engine off or on).	• The BCM constantly receives the elapsed time since the ignition was switched ON (engine off or on) sent via CAN communication from the instrument cluster. • If a DTC is detected, the BCM records the elapsed time since the ignition was switched ON (engine off or on) when the DTC was detected, and it is displayed in the Mazda Modular Diagnostic System (M-MDS).	—

Snapshot data item	Unit		Data contents	Data read/use method	Corresponding data monitor items
PWR_MODE_KEY	Key Out/Key Recently Out (Position 0)/ Accessory (Position 1)/Post Ignition (Position 2)/Ignition On (Position 2)/ Running (Position 2)/Running - Starting		• Key Out: Ignition switched off • Key Recently Out (Position 0): Elapsed time **within 3 s** since ignition was switched off • Accessory (Position 1): Ignition is switched to ACC • Post Ignition (Position 2): Elapsed time **within 3 s** since ignition was switched ON (engine off or on) • Ignition On (Position 2): Ignition switched ON (engine off) • Running (Position 2): Ignition switched ON (engine on) • Running - Starting: Cranking condition	• The BCM constantly receives the ignition switch status sent via CAN communication from the instrument cluster. • If a DTC is detected, the BCM records the ignition switch status when the DTC was detected, and it is displayed in the Mazda Modular Diagnostic System (M-MDS).	—
RPM_STATUS	Engine Stop/ Under1500rpm/ Over1500rpm/FAIL		Engine speed status	• The BCM constantly receives the engine speed sent via CAN communication from the instrument cluster. • If a DTC is detected, the BCM records the engine speed when the DTC was detected, and it is displayed in the Mazda Modular Diagnostic System (M-MDS).	TACHOMTR*1
SHIFT_STATUS	P/N/D/R/FAIL		Selector lever position status	• The BCM constantly receives the selector lever position sent via CAN communication from the instrument cluster. • If a DTC is detected, the BCM records the selector lever position when the DTC was detected, and it is displayed in the Mazda Modular Diagnostic System (M-MDS).	—
TOTAL_DIST	km	Miles	Accumulated total traveled distance from completion of vehicle until BCM detects DTC (Odometer value in instrument cluster)	The total traveled distance from which the BCM detects DTCs to the present can be calculated by performing the following procedure. 1. Verify the odometer value in the instrument cluster. 2. Verify the snapshot data item TOTAL_DIST. 3. Subtract 2 from 1.	—

Snapshot data item	Unit	Data contents	Data read/use method	Corresponding data monitor items
TOTAL_TIME	hh:mm:ss*2	Accumulated total elapsed time since vehicle completion until BCM detects a DTC **Note** • When the ROOM fuse is removed, and the ignition is switched off, the time is not included in the elapsed time.	The elapsed time from which the BCM detects DTCs to the present can be calculated by performing the following procedure. 1. Verify the instrument cluster PID item TOTAL_TIME. 2. Verify the snapshot data item TOTAL_TIME. 3. Subtract 2 from 1.	TOTAL_TIME*1
VPWR	V	BCM power supply voltage	—	VPWR_B
VSPD_STATUS	Stop/0-10km/h/ Over10km/h/FAIL	Vehicle speed status	• The BCM constantly receives the vehicle speed sent via CAN communication from the instrument cluster. • If a DTC is detected, the BCM records the vehicle speed when the DTC was detected, and it is displayed in the Mazda Modular Diagnostic System (M-MDS).	SPEEDOMTR*1

*1 : Instrument cluster PID (See ON-BOARD DIAGNOSTIC [INSTRUMENT CLUSTER].)
*2 : The seconds may be indicated after the decimal point.

Data Monitor Function
- With the PID/data monitor function, the input/output signal monitor items set in the BCM can be selected and read out in real-time.
- The items which can be selected are as follows.

Note
- The [Type] in the PID/data monitor table is indicated as follows.
  - — Input: Value input to module
  - — Conversion: Value converted into value input to module
  - — Calculation: Value calculated for several input values in module
  - — Output: Value output by module

—: Not applicable

Item	Unit	Type	Content	Operation status (reference)	Terminal
ABK_HD_CS_R	If an accessory part is installed, it cannot be used.				
ABK_HN_CS_R	If an accessory part is installed, it cannot be used.				
BG_HORN_RLY	If an accessory part is installed, it cannot be used.				
BLWR_RLY	Off/On	Output	Displays the blower relay operation status.	• Off: Blower relay is off (ignition is switched off) • On: Blower relay is on (ignition is switched ON)	3E
BRAKE_SW	Off/On	Conversion	Displays the brake switch (No.1 signal) operation status.	• Off: Brake switch (No. 1 signal) is off (brake pedal is not depressed) • On: Brake switch (No. 1 signal) is on (brake pedal is depressed)	2Z

Item	Unit	Type	Content	Operation status (reference)	Terminal
BRK_F_L_SW	Off(Normal)/On (Low)	Conversion	Displays the brake fluid level status.	• Off (Normal): Brake fluid level is between MAX and MIN • On (Low): Brake fluid level is lower than MIN	3R
C_DL_SW_LK	Off/On	Conversion	Displays the door lock switch operation status (lock operation).	• Off: door lock switch is not in lock • On: door lock switch is in lock	2K
C_DL_SW_UNL	Off/On	Conversion	Displays the door lock switch operation status (unlock operation).	• Off: door lock switch is not in unlock • On: door lock switch is in unlock	2K
DOOR_ALL	Close/OPEN	Conversion	Displays all the doors (front door, rear door) open/closed condition.	• Close: All doors are closed • OPEN: Any door is open	2H/2G/2F/2E
DOOR_D	Close/OPEN	Conversion	Displays the front door (RH) open/closed condition.	• Close: Front door (RH) is closed • OPEN: Front door (RH) is open	2H
DOOR_LR	Close/OPEN	Conversion	Displays the rear door (LH) open/closed condition.	• Close: Rear door (LH) is closed • OPEN: Rear door (LH) is open	2E
DOOR_P	Close/OPEN	Conversion	Displays the front door (LH) open/closed condition.	• Close: Front door (LH) is closed • OPEN: Front door (LH) is open	2G
DOOR_RR	Close/OPEN	Conversion	Displays the rear door (RH) open/closed condition.	• Close: Rear door (RH) is closed • OPEN: Rear door (RH) is open	2F
DR_LK_RY	Off/On	Output	Displays the door lock relay operation request (lock signal) send status. The BCM sends a lock signal to the ESU via LIN communication.	• Off: Lock signal is not sent (lock operation is not operated) • On: Lock signal is sent (lock operation is operated)	2Y
DR_UNLK_RY	Off/On	Output	Displays the door unlock relay operation request (unlock signal) send status. The BCM sends an unlock signal to the ESU via LIN communication.	• Off: Unlock signal is not sent (unlock operation is not operated) • On: Unlock signal is sent (unlock operation is performed)	2Y
F_FOG_RELAY	Off/On	Output	Displays the front fog light relay operation status.	• Off: Front fog light relay is off (fog light switch is in OFF position) • On: Front fog light relay is on (light switch is in TNS or HEAD position and fog light switch is in F.FOG position)	3I
F_FOG_SW	Off/On	Conversion	Displays the fog light switch operation status. The BCM receives a fog light switch input signal via CAN communication from the start stop unit.	• Off: Fog light switch OFF signal is received (fog light switch is in OFF position) • On: Fog light switch ON signal is received (fog light switch is in F.FOG position)	2C/2D
FUEL_SEN_HM	Displayed on Mazda Modular Diagnostic System (M-MDS) but does not function.				
FUEL_SEN_HS	Displayed on Mazda Modular Diagnostic System (M-MDS) but does not function.				
FUEL_SEN_HV	Displayed on Mazda Modular Diagnostic System (M-MDS) but does not function.				
FUEL_SEN_M	V	Input	Displays the fuel gauge sender unit input voltage. If the fuel tank level is low, the BCM input voltage increases because the fuel gauge sender unit resistance increases.	Displays the fuel gauge sender unit voltage.	2Q
FUEL_SEN_S	Displayed on Mazda Modular Diagnostic System (M-MDS) but does not function.				

Item	Unit	Type	Content	Operation status (reference)	Terminal
FUEL_SEN_V	V	Input	Displays the fuel gauge sender unit supply voltage.	Ignition switched ON (engine off): Approx. 4.15 V	1A/1B
H/L_CLN_RY	Off/On	Output	Displays the headlight cleaner relay operation status.	**Note** • For the specific operation condition of the PID [H/L_CLN_RY], refer to the [HEADLIGHT CLEANER SYSTEM] for new vehicle instruction. • Off: Headlight cleaner relay is off (headlight cleaner system is not operating) • On: Headlight cleaner relay is on (headlight cleaner system is operating)	3J
H/L_CS_RLS	Off/On	Conversion	Displays the headlight LO on request signal reception status. The BCM receives a headlight LO on request signal via LIN communication from the auto-light sensor/rain sensor.	**Note** • When the auto-light sensor/rain sensor detects the average 400 lux or less per 1 min for upward illumination, it sends a headlight LO on request signal to the BCM. • Off: Headlight LO on request signal is not received • On: Headlight LO on request signal is received	3X
H/L_DRL	Off/On	Output	Displays the running light on request signal send status. The BCM sends a running light on request signal to the ESU via LIN communication.	**Note** • For the specific operation condition of the PID [H/L_DRL], refer to the [RUNNING LIGHT SYSTEM] for new vehicle instruction. • Off: Running light on request signal is not sent • On: Running light on request signal is sent	2Y
H/L_HI	%	Output	Displays the headlight HI on request signal send status. The BCM sends a headlight HI on request signal to the ESU via LIN communication.	• Headlight HI on request signal is not sent (headlight high beam turns off): 0 % • Headlight HI on request signal is sent (headlight high beam turns on): 100 %	2Y
H/L_LOW	%	Output	Displays the headlight LO on request signal send status. The BCM sends a headlight LO on request signal to the ESU via LIN communication.	• Headlight LO on request signal is not sent (headlight low beam turns off): 0 % • Headlight LO on request signal is sent (headlight low beam turns on): 100 %	2Y
H/L_PASS	Off/On	Output	Displays the headlight LO and HI on request signal send status. The BCM sends a headlight LO and HI on request signal to the ESU via LIN communication.	• Off: Headlight LO and HI on request signal is not sent (no passing) • On: Headlight LO and HI on request signal is sent (passing)	2Y

Item	Unit	Type	Content	Operation status (reference)	Terminal
H/L_SW_HI	Off/On	Conversion	Displays the light switch (HI position) operation status. The BCM receives a light switch (HI position) input signal via CAN communication from the start stop unit.	• Off: Light switch (HI position) OFF signal is received • On: Light switch (HI position) ON signal is received	2C/2D
H/L_SW_LOW	Off/On	Conversion	Displays the light switch (LO position) operation status. The BCM receives a light switch (LO position) input signal via CAN communication from the start stop unit.	• Off: Light switch (LO position) off signal is received • On: Light switch (LO position) on signal is received	2C/2D
H/L_SW_OFF	Off/On	Conversion	Displays the light switch (OFF position) operation status. The BCM receives a light switch (OFF position) input signal via CAN communication from the start stop unit.	• Off: Light switch (OFF position) off signal is received • On: Light switch (OFF position) on signal is received	2C/2D
H/L_SW_PASS	Off/On	Conversion	Displays the light switch (passing position) operation status. The BCM receives a light switch (passing position) input signal via CAN communication from the start stop unit.	• Off: Light switch (passing position) off signal is received • On: Light switch (passing position) on signal is received	2C/2D
H/L_SW_TNS	Off/On	Conversion	Displays the light switch (TNS position) operation status. The BCM receives a light switch (TNS position) input signal via CAN communication from the start stop unit.	• Off: Light switch (TNS position) off signal is received • On: Light switch (TNS position) on signal is received	2C/2D
H/L_TNS	Off/Right-On/Left-On/On	Output	Displays the TNS on request signal send status. The BCM sends a TNS on request signal to the ESU via LIN communication.	• Off: TNS on request signal is not sent (light switch is in position other than TNS) • Right-On: Parking light (RH) is forcibly operated using simulation item • Left-On: Parking light (LH) is forcibly operated using simulation item • On: TNS on request signal is sent (light switch is in TNS position)	2Y
H/L_TNS_RLS	Off/On	Conversion	Displays the TNS on request signal reception status. The BCM receives a TNS on request signal via LIN communication from the auto-light sensor/rain sensor.	**Note** • When the auto-light sensor/rain sensor detects the average 700 lux or less per 1 min for upward illumination, it sends a TNS on request signal to the BCM. • Off: TNS on request signal is not received • On: TNS on request signal is received	3X

Item	Unit	Type	Content	Operation status (reference)	Terminal
H/L_TYPE	Halogen_2Lights/ Halogen_4Lights/ Lo-HID&Hi-Hlogen_4Lights/ LED_4Lights/Bi-Xenon	—	Displays the type of headlight equipped to the vehicle. The BCM sends information concerning the type of headlight to the ESU via LIN communication.	—	2Y
HAZARD_SW	Off/On/Unknown	Conversion	Displays the hazard warning operation status. The BCM receives a hazard warning switch input signal via CAN communication from the start stop unit.	• Off: Hazard warning switch off signal is received (hazard warning switch is not pressed) • On: Hazard warning switch on signal is received (hazard warning switch is pressed) • Unknown: Signal which hazard warning switch position is not determined is received	2C/2D
HOOD	Close/OPEN	Conversion	Displays the hood latch switch operation status.	• Close: Hood latch switch is on (hood is closed) • OPEN: Hood latch switch is off (hood is open)	3W
HORN_RLY	Off/On	Output	Displays the horn relay operation status.	• Off: Horn relay is off (horn switch is not pressed) • On: Horn relay is on (horn switch is pressed)	3H
IG1_STATUS	IG-Off/IG-On	Conversion	Displays the IG1 power supply input status determined in the BCM.	• IG-Off: IG1 power is not input to BCM • IG-On: IG1 power is input to BCM	3O
KC_SW_D_LK	Off/On	Conversion	Displays the front door key cylinder switch operation status (lock operation).	• Off: Front door key cylinder switch is not in lock (lock operation is not performed using front door key cylinder) • On: Front door key cylinder switch is in lock (lock operation is performed using front door key cylinder)	2L
KC_SW_D_UNL	Off/On	Conversion	Displays the front door key cylinder switch operation status (unlock operation).	• Off: Front door key cylinder switch is not in unlock (unlock operation is not performed using front door key cylinder) • On: Front door key cylinder switch is in unlock (unlock operation is performed using front door key cylinder)	2L
LL_SW_D_LK	Off/On	Conversion	Displays the front door lock-link switch operation status (lock operation).	• Off: Front door lock-link switch is not in lock (lock operation is not performed using driver's door lock knob) • On: Front door lock-link switch is in lock (lock operation is performed using driver's door lock knob)	2A
LL_SW_D_UNL	Off/On	Conversion	Displays the front door lock-link switch operation status (unlock operation).	• Off: Front door lock-link switch is not in unlock (unlock operation is not performed using driver's door lock knob) • On: Front door lock-link switch is in unlock (unlock operation is performed using driver's door lock knob)	2B

Item	Unit	Type	Content	Operation status (reference)	Terminal
LL_SW_P/R	Off/On	Conversion	Displays the door lock-link switch (passenger-side/rear-side) operation status (lock operation).	• Off: Door lock-link switch (passenger-side/rear-side) is not in lock (lock operation is not performed using door lock knob (passenger-side/rear-side)) • On: Door lock-link switch (passenger-side/rear-side) is in lock (lock operation is performed using door lock knob (passenger-side/rear-side))	2M
PNC_HD_CS_R	If an accessory part is installed, it cannot be used.				
PNC_HN_CS_R	If an accessory part is installed, it cannot be used.				
PTC_HEAT_ST	%	Output	Displays the PTC heater operation status (PTC heater output value controlled by BCM).	—	2AA
R_DEF_RLY	Off/On	Output	Displays the rear window defogger relay operation status.	• Off: Rear window defogger relay is off (rear window defogger switch is not pressed) • On: Rear window defogger relay is on (rear window defogger switch is pressed)	3K
R_FOG_LAMP	Off/On	Output	Displays the rear fog light on request signal send status. The BCM sends a rear fog light on request signal to the ESU via LIN communication.	• Off: Rear fog light on request signal is not sent (fog light switch is in OFF position) • On: Rear fog light on request signal is sent (fog light switch is in TNS or HEAD position and fog light switch is in R.FOG position)	2Y
R_FOG_SW	Off/On	Conversion	Displays the fog light switch operation status. The BCM receives a fog light switch input signal via CAN communication from the start stop unit.	• Off: Fog light switch OFF signal is received (fog light switch is in OFF position) • On: Fog light switch ON signal is received (fog light switch is in R.FOG position)	2C/2D
REVERSE_LMP	Off/On	Output	Displays the back-up light on request signal send status. The BCM sends a back-up light on request signal to the ESU via LIN communication.	• Off: Back-up light on request signal is not sent (MTX: shift lever is not in reverse position, ATX: selector lever is not in R position) • On: Back-up light on request signal is sent (MTX: shift lever is in reverse position, ATX: selector lever is in R position)	2Y
RLS_AAT	Displayed on Mazda Modular Diagnostic System (M-MDS) but does not function.				
RLS_WS_TEMP	Displayed on Mazda Modular Diagnostic System (M-MDS) but does not function.				

Item	Unit	Type	Content	Operation status (reference)	Terminal
ROOM_LAMP	Off/100%_On/60%_On/50%_On/40%_On/30%_On/20%_On	Output	Displays the front map light on request signal send status. The BCM sends a front map light on request signal to the ESU via LIN communication.	**Note** • For the specific operation condition of the PID [ROOM_LAMP], refer to the [INTERIOR LIGHT CONTROL SYSTEM] for new vehicle instruction. • Off: Front map light on request signal is not sent • 100%_On: Front map light on request signal (100%) is sent • 60%_On: Front map light on request signal (60%) is sent • 50%_On:— • 40%_On:— • 30%_On:— • 20%_On:—	2Y
SEAT_B_2C	Unbuckled/Buckled	Conversion	Displays the rear center buckle switch operation status.	• Unbuckled: Rear center buckle switch is off (rear seat belt (center) is not fastened) • Buckled: Rear center buckle switch is on (rear seat belt (center) is fastened)	2O
SEAT_B_2L	Unbuckled/Buckled	Conversion	Displays the rear buckle switch (LH) operation status.	• Unbuckled: Rear buckle switch (LH) is off (rear seat belt (LH) is not fastened) • Buckled: Rear buckle switch (LH) is on (rear seat belt (LH) is fastened)	2N
SEAT_B_2R	Unbuckled/Buckled	Conversion	Displays the rear buckle switch (RH) operation status.	• Unbuckled: Rear buckle switch (RH) is off (rear seat belt (RH) is not fastened) • Buckled: Rear buckle switch (RH) is on (rear seat belt (RH) is fastened)	2P
SEAT_B_3C	Displayed on Mazda Modular Diagnostic System (M-MDS) but does not function.				
SEAT_B_3L	Displayed on Mazda Modular Diagnostic System (M-MDS) but does not function.				
SEAT_B_3R	Displayed on Mazda Modular Diagnostic System (M-MDS) but does not function.				
STOP_LAMP_DSC	Unknown/Off/On/Backup_On	Conversion	Displays the brake light signal reception status. The BCM receives a brake light signal via CAN communication from the DSC HU/CM.	• Unknown: Brake light signal cannot be received • Off: Brake light signal (turn-off) is received (SCBS brake control is not operating) • On: Brake light signal (turn-on) is received (SCBS brake control is operating) • Backup_On: Brake light signal (turn-on) is received (malfunction is occurring in brake light switch)	3C/3D
STOP_LMP_CTL	Off/On	Output	Displays the brake light relay operation status.	**Note** • When the brake control (SCBS brake) is operating, the brake light relay turns on. • Off: Brake light relay is off (brake pedal is not depressed) • On: Brake light relay is on (brake pedal is depressed)	1D

Item	Unit	Type	Content	Operation status (reference)	Terminal
STOP_LMP_IN	Off/On	Conversion	Displays the brake light operation status.	• Off: Brake light turns off (brake pedal is not depressed) • On: Brake light turns on (brake pedal is depressed)	2AB
TR_UNL_SW	Off/On	Conversion	**4SD** • Displays the trunk lid opener switch (exterior) operation status. **5HB** • Displays the liftgate opener switch operation status.	**4SD** • Off: Trunk lid opener switch (exterior) is off (trunk lid opener switch (exterior) is not pressed) • On: Trunk lid opener switch (exterior) is on (trunk lid opener switch (exterior) is pressed) **5HB** • Off: Liftgate opener switch is off (liftgate opener switch is not pressed) • On: Liftgate opener switch is on (liftgate opener switch is pressed)	2S
TR_UNL_SW_IP	Off/On	Conversion	Displays the trunk lid opener switch (interior) operation status.	• Off: Trunk lid opener switch (interior) is off (trunk lid opener switch (interior) is not pressed) • On: Trunk lid opener switch (interior) is on (trunk lid opener switch (interior) is pressed)	3U
TRUNK	Close/OPEN	Conversion	**4SD** • Displays the trunk lid latch switch operation status. **5HB** • Displays the Liftgate latch switch operation status.	**4SD** • Close: Trunk lid latch switch is off (trunk is closed) • OPEN: Trunk lid latch switch is on (trunk is open) **5HB** • Close: Liftgate latch switch is off (liftgate is closed) • OPEN: Liftgate latch switch is on (liftgate is open)	2AC
TRUNK_LAMP	Off/On	Output	**4SD** • Displays the trunk compartment light on request signal send status. The BCM sends a trunk compartment light on request signal to the ESU via LIN communication. **5HB** • Displays the cargo compartment light on request signal send status. The BCM sends a cargo compartment light on request signal to the ESU via LIN communication.	**4SD** • Off: Trunk compartment light on request signal is not sent (trunk is closed) • On: Trunk compartment light on request signal is sent (trunk is open) **5HB** • Off: Cargo compartment light on request signal is not sent (liftgate is closed) • On: Cargo compartment light on request signal is sent (liftgate is open)	2Y

Item	Unit	Type	Content	Operation status (reference)	Terminal
TRUNK_MOTOR	Off/On	Output	**4SD** • Displays the trunk release signal send status. The BCM sends a trunk release signal to the ESU via LIN communication. **5HB** • Displays the liftgate release signal send status. The BCM sends a liftgate release signal to the ESU via LIN communication.	**4SD** • Off: Trunk release signal is not sent (trunk lid opener switch (interior) and trunk lid opener switch(exterior) is not pressed) • On: Trunk release signal is sent (trunk lid opener switch (interior) or trunk lid opener switch(exterior) is pressed) **5HB** • Off: Liftgate release signal is not sent (liftgate opener switch is not pressed) • On: Liftgate release signal is sent (liftgate opener switch is pressed)	2Y
TURN_L_L	Off/On	Output	Displays the turn light (LH) operation status.	• Off: Turn light (LH) turns off (turn switch is in OFF position) • On: Turn light (LH) flashes (turn switch is in LH position)	1G/3B
TURN_L_R	Off/On	Output	Displays the turn light (RH) operation status.	• Off: Turn light (RH) turns off (turn switch is in OFF position) • On: Turn light (RH) flashes (turn switch is in RH position)	1E/3A
TURN_SW	Off/Turn_R_On/ Turn_L_On/ Unknown	Conversion	Displays the turn switch operation status. The BCM receives a turn switch input signal via CAN communication from the start stop unit.	• Off: Turn switch off signal is received (turn switch is in OFF position) • Turn_R_On: Turn switch RH signal is received (turn switch is in RH position) • Turn_L_On: Turn switch LH signal is received (turn switch is in LH position) • Unknown: Signal which turn switch position is not determined is received	2C/2D
VPWR	V	Input	Displays the BCM supply voltage.	Switch the ignition ON (engine off):Approx. 12 V	1A
VPWR_B_H	V	Input	Displays the BCM supply voltage.	Switch the ignition ON (engine off):Approx. 12 V	1A
VSPD	KPH	Output	Displays the vehicle speed. The BCM sends a vehicle speed signal to the auto-light sensor/rain sensor via LIN communication.	—	3X
WASHER_CTRL	Off/Front/Rear	Output	Displays the washer operation request signal send status. The BCM sends a washer operation request signal to the ESU via LIN communication.	• Off: Washer operation request signal is not sent (washer is not operating) • Front: Front washer operation request signal is sent (front washer switch is in ON position) • Rear: Rear washer operation request signal is sent (rear washer switch is in rear washer position)	2Y
WASHER_F_SW	Off/On	Conversion	Displays the front washer switch operation status. The BCM receives a front washer switch input signal via CAN communication from the start stop unit.	• Off: Front washer switch off signal is received (front washer switch is in OFF position) • On: Front washer switch on signal is received (front washer switch is in ON position)	2C/2D

Item	Unit	Type	Content	Operation status (reference)	Terminal
WASHER_FLD	Off(Normal)/On (Low)	Conversion	Displays the washer fluid level status.	• Off (Normal): Washer fluid level is normal • On (Low): Washer fluid level is low	3Q
WASHER_R_SW	Off/On	Conversion	Displays the rear washer switch operation status. The BCM receives a rear washer switch input signal via CAN communication from the start stop unit.	• Off: Rear washer switch off signal is received (rear washer switch is in OFF position) • On: Rear washer switch on signal is received (rear washer switch is in rear washer position)	2C/2D
WIP_CS_RLS	Off/Single_Wipe/ SPEED_LOW/ SPEED_HI/Failure	Conversion	Displays the auto wiper control request signal reception status. The BCM receives an auto wiper control request signal via LIN communication from the auto-light sensor/rain sensor.	• Off: Off signal is received • Single_Wipe: Intermittent operation signal is received • SPEED_LOW: Continuous low operation signal is received • SPEED_HI: High speed signal is received • Failure: Auto wiper control request signal cannot be received	3X
WIP_F_AS	P_Posi/Not_P_Posi	Conversion	Displays the autostop switch operation status (windshield wiper arm and blade position).	• P_Posi: Windshield wipers are stopped in park position • Not_P_Posi: Windshield wipers are stopped in position other than park	3M
WIP_F_INT_L	%	Conversion	Displays the sensitivity adjustment volume operation status. The BCM receives a sensitivity adjustment volume input signal via CAN communication from the start stop unit.	The value changes according to the sensitivity adjustment volume operation.	2C/2D
WIP_F_RY	Off/On	Output	Displays the wiper INT relay operation status.	• Off: Wiper INT relay is off (windshield wiper are not operating) • On: Wiper INT relay is on (windshield wipers are operating)	3F
WIP_F_RY_HI	Off/On	Output	Displays the wiper HI relay operation status.	**Note** • If the windshield wiper switch is in the INT/AUTO position, the wiper HI relay turns on depending on the amount of rainfall. • Off: Wiper HI relay is off (windshield wiper switch is in position other than HI) • On: Wiper HI relay is on (windshield wiper switch is in HI position)	3G

Item	Unit	Type	Content	Operation status (reference)	Terminal
WIP_F_SW	OFF/LOW/HI/INT / AUTO	Conversion	Displays the windshield wiper switch operation status. The BCM receives a windshield wiper switch input signal via CAN communication from the start stop unit.	• Off: Windshield wiper switch off signal is received (windshield wiper switch is in OFF position) • LOW: Windshield wiper switch LOW signal is received (windshield wiper switch is in LO position) • HI: Windshield wiper switch HI signal is received (windshield wiper switch is in HI position) • INT/AUTO: Windshield wiper switch INT signal or AUTO signal is received (windshield wiper switch is in INT/AUTO position)	2C/2D
WIP_R_CTRL	Off/INT/On/ On_1Time/ On_2Time/ On_3Time/ On_4Time	Output	Displays the rear wiper operation request signal send status. The BCM sends a rear wiper operation request signal to the ESU via LIN communication.	• Off: Rear wiper operation request signal is not sent (rear wiper switch is in OFF position) • INT: Rear wiper INT operation request signal is sent (rear wiper switch is in INT position) • On: Rear wiper operation request signal is sent (rear wiper switch is in ON position) • On_1Time:— • On_2Time: Rear wiper INT operation request signal is sent (during synchronized washer operation) • On_3Time:— • On_4Time:—	2Y
WIP_R_INT	s	Output	Displays the rear wiper intermittent interval. The BCM sends a rear wiper intermittent interval to the ESU via LIN communication.	—	2Y
WIP_R_SW	OFF/LOW/INT	Conversion	Displays the rear wiper switch operation status. The BCM receives a rear wiper switch input signal via CAN communication from the start stop unit.	• Off: Rear wiper switch off signal is received (rear wiper switch is in OFF position) • LOW: Rear wiper switch LOW signal is received (rear wiper switch is in ON position) • INT: Rear wiper switch INT signal is received (rear wiper switch is in INT position)	2C/2D

Simulation function

- The simulation function drives the output parts set in the BCM regardless of the BCM control condition.
- The items which can be selected are as follows.

Note

- Simulation items on parts which are not equipped to the vehicle do not operate.

Simulation item	Unit/ Operation	Data contents	Output part name	Operation condition
BG_HORN_RLY	If an accessory part is installed, it cannot be used.			
BLWR_RLY	Off/On	• Off: Turns the blower relay off. • On: Turns the blower relay on.	Blower relay	Ignition switched ON (engine off or on)

Simulation item	Unit/ Operation	Data contents	Output part name	Operation condition
DR_LK_RY	Off/On	**Note** • The BCM sends a door lock relay operation request signal (lock signal) to the ESU via LIN communication. • Off: Does not lock the door. • On: Locks the door.	Door lock relay	Ignition switched ON (engine off or on)
DR_UNLK_RY	Off/On	**Note** • The BCM sends a door unlock relay operation request signal (unlock signal) to the ESU via LIN communication. • Off: Does not unlock the door. • On: Unlocks the door.	Door unlock relay	Ignition switched ON (engine off or on)
F_FOG_RELAY	Off/On	• Off: Turns the front fog light off. • On: Turns the front fog light on.	Front fog light relay	Ignition switched ON (engine off or on)
H/L_CLN_RY	Off/On	• Off: Turns the headlight cleaner relay off. • On: Turns the headlight cleaner relay on.	Headlight cleaner relay	Ignition switched ON (engine off or on)
H/L_DRL	Off/On	**Note** • The BCM sends a running light on request signal to the ESU via LIN communication. • Off: The running lights are turned off. • On: The running lights are turned on.	Running light	Ignition switched ON (engine off or on)
H/L_HI	0%/100%	**Note** • The BCM sends a headlight HI on request signal to the ESU via LIN communication. • 0%: The headlights HI are turned off. • 100%: The headlights HI are turned on.	Headlight HI	Ignition switched ON (engine off or on)
H/L_LOW	0%/100%	**Note** • The BCM sends a headlight LO on request signal to the ESU via LIN communication. • 0%: The headlights LO are turned off. • 100%: The headlights LO are turned on.	Headlight LO	Ignition switched ON (engine off or on)
H/L_TNS	Off/Right-On/Left-On/ On	**Note** • The BCM sends a TNS on request signal to the ESU via LIN communication. • Off: Turns the parking lights off. • Right-On: Turns the parking light (RH) on. • Left-On: Turns the parking light (LH) on. • On: Turns the parking lights (RH/LH) on.	Parking light	Ignition switched ON (engine off or on)
HORN_RLY	Off/On	• Off: Stops the horn. • On: Sounds the horn.	Horn relay	Ignition switched ON (engine off or on)

Simulation item	Unit/ Operation	Data contents	Output part name	Operation condition
PTC_HEAT_ST	0%/10%/ 20%/30%/ 40%/50%/ 60%/70%/ 80%	• 0%: Stops PTC heater output. • 10%: Controls PTC heater output level to 10%. • 20%: Controls PTC heater output level to 20%. • 30%: Controls PTC heater output level to 30%. • 40%: Controls PTC heater output level to 40%. • 50%: Controls PTC heater output level to 50%. • 60%: Controls PTC heater output level to 60%. • 70%: Controls PTC heater output level to 70%. • 80%: Controls PTC heater output level to 80%.	PTC heater	Ignition switched ON (engine off or on)
R_DEF_RLY	Off/On	• Off: Stops the rear window defogger. • On: Operates the rear window defogger.	Rear window defogger relay	Ignition switched ON (engine off or on)
R_FOG_LAMP	Off/On	• Off: Turns the rear fog light off. • On: Turns the rear fog light on.	Rear fog light relay	Ignition switched ON (engine off or on)
REVERSE_LMP	Off/On	**Note** • The BCM sends a back-up light on request signal to the ESU via LIN communication. • Off: Turns the back-up light off. • On: Turns the back-up light on.	Back-up light	Ignition switched ON (engine off or on)
ROOM_LAMP	Off/100% _On	**Note** • The BCM sends a front map light on request signal to the ESU via LIN communication. • Off: Turns the front map light off. • 100%_On: Turn the front map light on.	Front map light	Ignition switched ON (engine off or on)
STOP_LMP_CTL	Off/On	• Off: Turns the brake light off. • On: Turns the brake light on.	Brake light relay	Ignition switched ON (engine off or on)
TRUNK_LAMP	Off/On	**Note** • The BCM sends a cargo compartment light on request signal to the ESU via LIN communication. • Off: Turns the cargo compartment light off. • On: Turns the cargo compartment light on.	Cargo compartment light	Ignition switched ON (engine off or on)
TRUNK_MOTOR	Off/On	**Note** • The BCM sends a liftgate release signal to the ESU via LIN communication. • Off: Does not open the liftgate. • On: Opens the liftgate.	Liftgate release relay	Ignition switched ON (engine off or on)

Simulation item	Unit/ Operation	Data contents	Output part name	Operation condition
WASHER_CTRL	Off/ Front_ON	**Note** • The BCM sends a front washer operation request signal to the ESU via LIN communication. • Off: Stops the windshield washer motor. • Front_ON: Operates the windshield washer motor.	Windshield washer motor	Ignition switched ON (engine off or on)
WIP_R_CTRL	Off/INT/On/ On_1Time/ On_2Time/ On_3Time/ On_4Time	**Note** • The BCM sends a rear wiper operation request signal or rear wiper INT operation request signal to the ESU via LIN communication. • Off: Stops the rear wiper. • INT: Operates the rear wiper intermittently. • On: Operates the rear wiper continuously. • On_1Time: — • On_2Time: Operates the rear wiper twice. • On_3Time: — • On_4Time: —	Rear wiper motor	Ignition switched ON (engine off or on)